1

Question:

Scale and its types. Explain different kinds of rating scale by giving at least two examples of each.

Ans:

Scale:

A scale is a set of symbols or numbers that can be assigned by a rule to the individuals. The four types of scales are: ~~nominal~~ nominal, ordinal, interval and ratio.

Nominal Scale:

A nominal scale is the classification of individuals, objects or responses having common characteristics. A variable measured on a nominal scale may have more than one category depending upon the extent of variation.

Examples:

Sex: Male, Female

Area: Rural, Urban, Suburban

Ordinal Scale:

It has the characteristics of nominal scale plus an ability to order data. It ranks subgroups in a certain order. They are arranged either in ascending or descending order.

Example:

Income can be measured either quantitatively (in Rupees) or qualitatively using subgroups 'above average', 'average' and 'below average'.

Interval Scale:

Contains all the features of ordinal scale with the added dimension that intervals b/w the points on the scale are equal. It has starting and terminating points that is divided into equally units intervals.

Example:

Celsius System - Starting point is zero and terminating point is 100°c

Ratio Scale:

It has all the properties of nominal, ordinal and interval scale plus its own property: the zero point of a ratio scale is fixed. It is an absolute scale the difference b/w the intervals is always measured from zero point. The measurement of income, age, hieght and weight are examples of this scale. A person who is 40 years of age is twice as old as a 20 years old.

Types Of Rating Scale:

1- Dichotomous Scale:

The dichotomous scale is used to elicit a Yes or No answer.

Example:

Do you own a car? Yes, No

2- Category Scale:

The category scale uses multiple items to elicit a single response.

Example:

Where is Northern California do you resided?

North Bay – East Bay – South Bay – Peninsula – Others

3- Likert Scale:

The likert scale is designed to examine how strongly subject agree or disagree with statement

Example:

Strongly Disagree Disagree Neutral Agree Strongly disagree

4- Semantic Differential Scale:

Several bipolar attributes are identified at the extremes of the scale and respondents are asked to indicate their attitude. The bipolar attributes are Good – Bad, Strong – Weak, Hot – Cold. It is used to assess respondents attitude towards a partical brand.

Example:

Responsive ______ Unresponsive

Beautiful ______ Ugly

5- Numerical Scale:

The numerical scale is similar to semantic differential scale with the difference that numbers on a 5-point or 7-point scale are provided with bipolar attributes at both ends

Example:

Extremely Pleased 7 6 5 4 3 2 1 Extremely displeased

6- Itemized Rating Scale:

A 5-Point or 7-Point Scale with attributes is provided for each item and circles the relevent number against each item

Example:

Very unlikely	Unlikely	Neutral	Likely	Very likely	
1	2	3	4	5	(Balanced)

OR

Not all interested	Somewhat Interested	Moderately Interested	Very much Interested	
1	2	3	4	(unbalanced)

A scale which have a neutral point is called Balanced Rating Scale. A scale which does not have a neutral point is called unbalanced Rating Scale.

7- Graphic Rating Scale:

A Graphical representation helps the respondents to answer a particular question by placing mark at the appropriate point on line

10 Excelent

5 all right

1 very bad

Question:
What is data Collection? Explain main sources for data Collection.

Ans:

Data Collection:
Data Collection is a term used to describe a process of preparing and collecting data the purpose of data collection is to obtain information to Keep on record, to make decision about important issues, to pass information on to others.

Data Can be Classified as
(i) Primary data
(ii) Secondary data

Primary Data:
The data that are collected from the field under the control and supervision of an investigator is known as primary data.

Example:
Data Collected using the survey to determine the market segment of any product.

Sources of Primary data:
There are four main primary sources of data

(i) Individuals:
Individuals are those who provide information when interviewed, administered, questionaire and observed.

(ii) Focus Group:
It is a qualitative research

technique in which people are informally interviewed in a group discussion setting a group of 8-10 members are led by a moderator in discussion on a one particular topic. It intends to find out how the people feel about a particular issue.

(iii) Panels:

Unlike the focus group, panels of members meet more than once in a group discussion setting. In cases, where the effect of certain changes are to be studied over a period of time, panel studied are very useful.

(iv) Unobtrusive measures:

These are the sources that does not involve people. For example, journals in a library indicate their popularity, frequency of use. The no of different brands in bags provide a measure of their consumption level.

Sources for Secondry data

Secondry data:

Secondry data are collected from sources which have been already created for the purpose of first time use and future uses.

Following are the main sources:

(i) Government or Semi Govt Publication:

There are many Govt or Semi-Govt organisations that collect data

on a regular basis and publish it for use of public and interest group members.
For examples
Census, Labor force survey, annual report of companies etc.

(ii) Earlier Research:
For some topics, an enormous number of research studies that have already been done by others can provide you with the required information.

(iii) Personal Records:
Some people write historical and personal records that may provide the information you need.

(iv) Mass Media:
Reports published in newspapers, magzines provides a good source for secondary data collection.

Question:
Define Research. Explain the difference b/w basic and applied research. (Describe the steps in the research process)

Ans:

Research:

Research is a systematic, formal and precise process employed to gain solutions to problem or to discover and interpret new facts and relationships. It is a process for collecting analyzing and interpreting information to answer questions relating to a phenomena.

Basic Research:

Basic Research is the source of new scientific ideas and ways of thinking about the world. It is conducted to expand knowledge, to verify the acceptability of given theory or learn more about the certain concept.

Applied Research:

Research conducted when a decision must be made about a real life problem. It relies on quick, small-scale study that provide practical results that could be used to solve a problem.

Steps in research process

- Choose topic
- Focus Research Question
- Design Study
- Collect data
- Analyze data
- Interpret data
- Inform others

Difference b/w Basic and Applied Research

Basic	Applied
1- Research is intrinsically satisfying and judgemental procedure.	1- Research is a part of job and is judged by others.
2- Research problems and subjects are selected with a great deal of freedom	2- Research Problems are "namely contrained" to the demand of others.
3- The main goal is to contribute to basic theoretical knowledge	3- The main goal is to have practical payoffs or uses of results.
4- The primary concern is with the internal logic and rigor of research design	4- The primary concern is with the ability to generalize findings to area of interests to sponsors.

Question:
What is Literature Review

Ans: Definition:
In term of the literature review "the literature" means the work you consult in order to understand and investigate your research problems. It is also of critical summary of previous research on the topic. It is an objective, throug summary and critical analysis of research literature on the topic being studied.

Importance / Purpose / Aim of the review

The purpose of the literature review is to set the stage for own research. The purpose of the literature review is to convey to the reader what knowledge and ideas have been established on a topic and what are the strenghts and weeknesses.

→ To what has and has not been investigated.
→ To develop general explanation for observed variations in a behavior or phenomenon.
→ Discovering important variables relevent to the topic.
→ To identify data sources that other researches have used.
→ To develop alternate research projects.
→ To discover how a research project is releated to the work of others.
→ Identify new ways to interpret, and shed light on any gaps in previous research.
→ Illustrate the importance of the topic.
→ To limit the problem.
→ Point the way forward for further research.

Procedure For Reviewing literature:

The whole process of reviewing includes
→ Searching for literature
→ Sorting the retrieved literature
→ Analytical reading of papers
→ Evaluating reading of papers
→ Comparison across studies
→ Organizing the content
→ Writing the review

Function of literature Review

The major functions are

1- To provide theoretical background to your /study /survey /research.

2- It is also help you to comparision your findings in relation to existing work.

3- To bring clearity and focus to your study

4- To improve your research methodology.

5- To broaden your knowledge.

Question:-

What is Pretesting Of a Questionnaire

Ans: The pretest provides means of catching and solving unforeseen problems in the use of the questionnaire, such as the phrasing and sequencing of questions.

To test the questionnaire that whether it will achieve the desire result, it is necessary to pretest the Questionnaire before it is used in a full scale survey.

Pretesting process seeks to determine whether respondents have any difficulty in understanding the questionnaire and whether there are any biased question.

Usually a small number of respondents are selected for the Pre-test.

Aims of Pretesting the Questionnaire

- Improve the wording.
- Correct and improve translation of technical term.
- Check the accuracy and adequacy of the questionnaires.

- Eliminate unnecessary questions and add necessary ones.
- Estimate the time needed to conduct the interview.
- To access whether the questions are understood by all classes of respondent.
- Access whether the questions have been placed in the best order.

Advantages of Pretesting of questionnaire

1- It can determine the strengths and weekness of your survey concerning question format, working and order.
2- It is very useful method for determine whether respondents have any difficulty understanding the questionnaire.
3- It ensures that there is costly error in questionnaire which are given to a large no of respondent avoided.
4- We can also be sure about questionnaire format by it.

Disadvantages of Pretesting of questionnaire

1- Its application is limited to a study population that can read and write. It cannot be used on a population.
2- Questionnaire are notorious for their low response rate; e.g people fail to return them.
3- If respondents do not understand some questions there is no opportunity for them to have the meaning clarified.
4- Mailed questionnaire are inappropriate when spontaneous responses are required.

What do You Know about "Questionnaire designing"
A tool for collecting information to describe, compare or explain knowledge, attitudes, behaviour and/or socio demographic characteristics on a particular target group. Each stage of research process is important because of its interdependence with other stages of the process. How a survey is only as good as the question it asks. While good grammer and common sense are important in question writing, more is required in the art of questionnaire design.

Discuss the qualities of well-designed Questionnaire
Relevancy and accuracy are the two basic criteria a questionnaire must meet if it is to be achieve the researcher purposes.

(i) Questionnaire relevancy:
A questionnaire is relevant if no unnecessary information is collected and if the information that is needed to solve the problem is obtained. Asking the wrong or an irrelevant question is a pitfall to be avoided.

(ii) Questionnaire accuracy:
Accuracy means that the information is reliable and valid. While experienced researchers generally believe that one should use simple, understandable, unbiased, unambiguous words to ensure accuracy in question writing can be generalized across projects.

Factors to be taken into account:
1- What should be asked (content)
2- How should each question be phrased (wording)
3- In what sequence should the questions be arranged (sequence)
4- What questionnaire lay-out would best serve the research objective (lay out)
5- How should the questionnaire be pre tested?

1- Question Content:
Questions included in a questionnaire should be
- relevent & accurate
- Complete and focused
- Precise
- Balanced in terms of General Vs specific
- Multiple questions be favored to avoid double barrel questions.
- Participants information level should be understood
- Participants recall ability should be kept in mind

2- Question Wording:
The following principles of question wording in questionnaire design are
- Questions should contain shared vocabulary which is common to both parties.
- Unsupported assumptions should be avoided
- Frame of reference needs to be specific to avoid misinterpretations.
- Personalizing questions changes responses
- Avoid leading or loaded questions

- Avoid Complexity
- Avoid burdensome questions that may tax the respondent's memory
- Biased Wording should be avoided.

3- Question Sequence

- Start with simple and interesting questions
- Personal information towards the end
- Order bias can distort survey results
- Techniques to avoid order bias
  - → Funnel Technique
  - → Filter Questions
  - → Pivot Questions

4- Layout of Questionnaire

The physical appearance of a questionnaire can have a significant effect upon both the quantity and quality of data obtain. The use of booklets, in the place of loose or stapled sheets of paper makes it easier for interviewer or respondent to progress through the document.

Simple, Clear formats should be used. The clarity of questionnaire presentation can also help to improve the ease with which interviewers or respondents are able to complete a questionnaire. Creative use of space must be considered by the researcher.

Special Reference to Mail Questionnaire:

A mailed questionnaire must be accompanied by covering letter. One of the major problems with this method is the low response rate. In this case, the findings have very limited applicability to the pop study.

Q:

What is a question?

Ans: A sentence in interrogative form addressed to someone in order to get information in reply is called question

Example:

Here are some examples of sentences

- Tell me your name
- Give me your address

Explain Open ended and Closed ended question in detail

In an questionnaire, questions may be formulated as

1- Open Ended Question
2- Closed Ended Question

1- Open Ended Question :-

In an open-ended question the possible responses are not given. In the case of questionnaire, the respondent writes down the answers in his/her words, whereas in the case of an interview schedule the investigator records the answers in a summary discribing a respondent's answers.

Advantanges:

1- It can be used when interviewed has prior knowledge about a topic.
2- It provide respondents a chance to enlist their strong opinion.
3- It is used when immediatly follow a closed ended question.

Disadvantages:-

1- It can be very demanding for respondent e.g in questionnaire, some respondents may not be able to express themselves. So information can be lost.

2- It produce many different responses and few select some option (Low decision making)

3- It provide rarely accurate measurement

4- It requires enormous amount of time for data entry.

Closed Ended Question:-

In closed ended, The possible answers are set out in the questionnaire and the respondent ticks the category that best describes the respondent's answer. It is usually wise to provide a category "other/Please explain" to accommodate any response listed e.g

Q: What is your average Annual income?

Under 10,000 ☐
10,000 - 19,999 ☐
20,000 - 29,999 ☐
30,000 - 39,999 ☐
above 40,000 ☐

The questions in above figures are classified as close ended question.

Advantages:

1- It helps to ensure that the information needed by the researcher is obtained.

2- The case of answering a ready-made list of responses may creat tendency among some respondents to tick. Category without thinking through issue

3- As the possible responses are already categorised. They are easy to analyse.

Disadvantages:-

1- The information obtained through them lacks depth and Variety.

2- In Questionnaire the given response pattren for a question could condition the thinking of respondents and so the answers provided may not truly reflect respondents opinions.

Example for Open-ended Question:

A- What is your current age? ______ Years.

B- How would you describe your current marital?

C- What is your Annual income?

Question:-
Explain Validity and its types?
Ans:
Defination:-
In term of measurement procedure. Therefore, Validity is defined as the degree to which the researcher has measured what he has set out to measure.
Concept of Validity:
We take example to examine the concept of validity. Suppose that most of the questions in the interview schedule that your aims was to find out about health needs. But what attitudes respondents have to the health services thus the instrument is not measuring what it was designed to measure.
The defination arise some questions:
(i) who decides (that) that an instrument is measuring what is (the) supposed to me measure?
Obviously the answer to the question is the person who has designed the study and expert in the field.
There are three types of Validity
(1) Face and Content Validity:
(2) Concurrent and Predictive Validity
(3) Construct Validity
(1) Face and Content Validity:-
The Judgement that an instrument is measuring what it is based upon

the logical link b/w the question(s) and the objectives of the study. Advantages of this type is that it is easy to apply. Each question on the scale must have a logical link with an objective. This link is called Face Validity. Assessment of the question(s) of an instrument in this respect is called Content Validity.

2) Concurrent and Predictive Validity:

Predictive validity is judged by the degree to which an instrument can forecast an outcome.

Concurrent validity is judged by how well an instrument compares with second assessment concurrently done.

Correlation coefficient b/w the predicted status and criterion is usually possible to express predictive validity. Such coefficient is called validity coefficient.

3) Construct Validity:

This type is based upon statistical procedures. It is determined by verifying the contribution of each construct to the total variance observed in a phenomenon. Where total variance indicates the degree of validity of the instrument. The greater variance to the constructs is the higher the validity of the instrument. The disadvantage of this type is that you need to know about the required statistical procedures.

Question:-
What's the difference b/w 'References'
and 'Bibliograph'.
Ans: The term 'References' and 'Bibliography'
are often used synonymously, but there is
a difference in meaning between them.

References:-
References are the items you have read and
specifically referred to in your assignment
and your list of sources at the end of
the assignment will be headed 'References'.
If you make a point of reading selectively
you can make use of everything you read
by referring directly to it in your
assignments.
This enables them to quickly find in your
list of references that source you cited
and if necessary, check the validity of
it for themselves.

Bibliography:-
Strickly speaking, a bibliography is a list
of everything you read in preparation for
writing an assignment, whether or not
you referred specifically to it in the
assignment.
A bibliography will therefore normally contain
sources that you have cited in your assignment
and also those you found to be influential
but decided not to cite. A bibliography

Can give a tutor an overview of which authors have influenced your ideas and arguments even if you do not specifically refer to them.

Question:-
What are the principles of Sampling theory.

Ans:- The main aim of Sampling theory is to make sampling more efficient. So that the answer to a particular question is given in a valid, efficient and economical way. The theory of sampling is based on three important basic principles.

i- Principle of Validity
ii- Principle of Statistical Regularity
iii- Principle of Optimization

i- Principle Of Validity:-
This principle states that the sampling design provides valid estimates about population parameters. By valid we mean that the sample should be so selected that the estimates could be interpreted objectively and in terms of probability.

ii- Principle Of Statistical Regularity:-
The principle of Statistical Regularity, which has its origin in the theory of Probability.

can be explained in these words.
"A moderately large number of items chosen at random from a large group are almost sure on the average to process the characteristics of the large group"
This principle stress upon the desirability and importance of selecting sample designs where inclusion of sampling units in sample is based on probability theory.

iii- **Principle of Optimization:-**
Optimization is meant to develop method of sample selection and of estimation that provides
a- A given level of efficiency with the minimum possible resources
b- A given value of cost with the maximum possible efficiency

Thus the principle of Optimization minimizes the risk or loss of sampling design i.e the principle stresses upon obtaining optimum results with minimization of the total loss in terms of cost and Mean Square Error.

Question:-
What are the major sections involved in writting a research report?

Ans:- A proper research report includes the following sections, submitted in the order listed, each section to start on a new page.

- Style
- Title Page
- Abstract
- Introduction
- Materials and Methods
- Results
- Discussion
- Literature Cited

o Style :-

In all sections of your report, use paragraphs to seperate each important point, and present your points in logical order. Use present tense to report background that is already established.

For example,

'The grass is green'.

o Title Page :-

Select an informative title, such as "Role of temprature in detesmination of the rate of development of Xenopus larvae". A title such as "Biology lab # 1" is not informative. Include the name(s) and address(es) of all authors and data submitted.

O Abstract:-

Summarize the study, focusing on the results and major conclusions, including relevant quantitative data. It must be a single paragraph, and concise. It should stand on its own, therefore donot refer to any other part of the report, such as a figure or table. Avoid long sections of introductory. As a summary of work done, it is written is past tense.

O Introduction:-

Introduce the rationale behind the study, including

→ The overall question and its relevent to science

→ Suitability of the experimental model to the overall question

→ Experimental design and specific hypothesis

→ Significance of the anticipated results to the overall question

Include appropriate background information.

O Methods and Materials:-

The purpose of this section is to document all of your procedures so that another scientist could reproduce all or part of your work. It is not designed to be a set of instructions. It is standard practice to report methods and materials in past tense, third person passive.

Your laboratory notebook should contain all of the details of everything you do in lab, plus any additional information needed in order to complete this section.

o Results:-

Raw data are never included in a research paper. Analyze your data, then present the analyzed data in the form of a graph, table or in narrative form. Present the same data only once, in the most effective manner. By presenting converted data, you make your point succinctly and clearly.

o Discussion:-

Interpret your data in the discussion. Decide if each hypothesis is supported, rejected, or if you cannot make a decision with confidence. Do not simply dismiss a study or part of a study as "Inconclusive". Make what conclusions you can, the suggest how the experiment must be modified in order to properly test the hypothesis(es). Explain all of your observations as much as possible, focusing on mehanisms.

o Literature Cited:-

List all literature cited in your report, in alphabetical order, by first author. In a proper research paper, only primary literature is used. Some of your reports may not require references, and if that is the case simply state "No references were consulted".

Question:-
Proposal assignment (Research) requires these sections.

Ans:-
→ Summary
→ Problem definition
→ Scope of Project
→ Solution Criteria
→ Proposed Procedure
→ Schedule
→ Budget
→ Qualifications
→ Source of Information

o Summary:-
Summary is a miniature version of the whole proposal.
> Give brief overview of this information:
- Problem
- Scope and procedure
- Alternative Solutions
- Evaluation Criteria
- Budget for study
- Deliverables

- Problem Definition
  - > Describe problem or need
  - Give more background that in summary
  - Support with specific evidence
  - State from Clients point of views
- Scope of Project
  - > State topics to be investigated and how
  - > Project time needed to complete project
  - > List deliverables
- Evaluation Criteria
  - > Expand on listing of criteria in scope section
  - > Define criteria very specifically
  - > State how you will determine or measure for each criteria
- Solution Criteria

The alternative solutions will be evaluated according to the criteria. These criteria are listed in order of priority, from highest to lowest, as designated by the client.

- Proposed Procedure
  - > Breakdown your scope of work into tasks
  - > Include your methods of research and methods of evaluation
  - > Show sequence of tasks
  - > Include specific sources of information
  - > This is the narrative of your schedule
- Schedule
  - > Graphically present your specific tasks along a timeline.

> Include specific dates
> Breakdown tasks as much as possible

o Budget:-

> Present the costs for completing the study
> Include items such as research, estimating, testing, traveling, Photocopying, Phoning
> Remember, you are a consultant make clear your rate per hour for various tasks.
> Present budget graphically, but introduce with at least one sentence
> Label the figure

o Qualifications:-

> Present your qualifications for doing the study
> Evidence of qualifications includes these items

- Relevent courses taken
- Relevent employment
- Degree program in progress
- Any relevent experience

> Don't worry about not having lots of Experience

o Sources of Information:-

> This section is more for me than for a "real world" engineering boss.
> Include all relevant articles, books, people, reports, class notes, etc. From which you will get information relevent to your study.
> Be sure also to weave discussion of sources of information into proposed procedure
> Give this information for each source

- Name of author
- titles of Journals
- Titles of books.

Question:-
Explain the types of research on the basis of Purpos.

Ans:-
Following are the types on the basis of Purpose

1- Exploratory Research
2- Descriptive Research
3- Causal Research
4- Correlation Research

1- Exploratory Research

- Exploratory research is usually conducted when the researcher does not know much about the problem and needs additional information or desires new or more recent information.
- Given its fundamental nature, exploratory research often concludes that a perceived problem does not actually exist.

Goals

Exploratory research is used in a number of situations

→ To gain background information
→ To define terms
→ To clarify problems and hypothesis
→ To establish research priorities
→ Diagnose a situation
→ Discover new ideas
→ Screening of alternatives

Example:-

Suppose a Chinese fast food restaurant chain is expanding considering its hour and product line with a breakfast menu. Exploratory Research with a small number of current customers might find a strong negative reaction to eating a spicy vegetable breakfast at a chinese fast food outlet. Thus exploratory research might help crystallize a problem and identify information needed for future research.

2- Descriptive Research

Descriptive research, also known as statistical research, describes data and characteristics about the population or phenomenon being studied. Descriptive research answers the questions who, what, where, when and how.

Goals:-

→ Provide a detailed, highly accurate picture.
→ Locate new data that contradict past data.
→ Create a set of categories or classify types.
→ Clarify a sequence of steps or stages.

Example:-

For years TEEN magazine sensed that 12-15 years old girls cared lot about fragrances, lipsticks & mascaras but they lacked any quantitative evidence. The descriptive research found that 94.2% of 12-15 years old girls use cream, 86.4% use fragrance and 84.9% use lip-gloss.

3- Causal Research

Causal research deals with cause-and-effect relationship. Causal research involves experiment, where an independent variable is manipulated to see how it affect a dependent variable by controlling the effects of extraneous variable.

Example

A researcher may wish to compare the body composition. In this case the researcher is not manipulating any variable. Researcher only investigating the effect of free weights versus exercise machines on body composition. Obviously since other factors such as diet, training program, aerobic conditioning could effect body composition. Causal research must be reviewed carefully to see how these other factors were controlled.

4- Correlation Research

Correlation research investigates, the extent to which variations in one factor corresponds with variations in one or more other factors based on correlation coefficient. It is appropriate where variables are complex

Example:

The researcher's examines cigarette smoking and a variety of lungs disease. These two variables smoking and lung disease were found to co-vary together. Smoking, drinking and tobacco chewing associate with lungs disease is a correlation study.

Question:-

Define Error and its types, give some examples of sources of error involved when information are collected in a sample Surveys.

Ans:-

Error:-

A sample being only a part of a population can not perfectly represent the population no metter how much carefully the sample is selected. This result in difference b/w the values of sample statistic and the true value of the corresponding population parameter. Such a difference is called Error.

Types of Error:-

In sample Survey, the following types of four errors occur. Sample surveys yield accurate when researchers succeed in avoiding these kinds of errors.

1- Coverage Error:-

An error in coverage occurs when sampling frame does not include all elements of the population that researcher wish to study. Sampling frame is a list of person from which a sample is drawn. It means of which incomplete list introduce coverage error the survey designers are unaware coverage error may also occur in field procedure.

Example:

In Las Vegas, Nevada, a crime prevention project used a telephone directory to obtain a

Sample of names and adress for mail survey. Unfortunately, the survey did not produce accurate result because more than half of the residents of las Vegas have unlisted telephone numbers.

Sources/Problems leading to Coverage Error

Missing element: Some population elements are not include in the frame.

Clustering: Some lists refer to groups of elements, not including elements.

Ineligible Units: When some listings do not relate to elements of the survey population.

Duplicate listings: Some population elements have more than one listing.

How to avoid/Minimizing Coverage Error

→ Make any reasonable effort to minimize the coverage error by using the best sampling frame.

→ Every member of the population that the researcher is trying to decide would have an equal chance of being selected for the sample

→ Refine the target population to fit the frame better.

2- Measurement Error

Difference b/w actual value of the quantity and the value obtained by a measurement is called measurement error.

Measurement errors occurs when a respondent's answer to a given question is inaccurate imprecise or can't be compared in any useful way to other respondents answer. Measurement errors are made when we collect the data, not when we select the sample.

The size of measurement is the difference b/w the respondent's answer to a particular question and the correct answer.

Sources:

→ The Survey Method

→ The Questionnaire

→ The Interviewer

→ The respondent

Example:

A local council of ministries ~~Yes~~ sponsored a survey of community residents to learn more about their participation in religious activities. Included in the survey was the following question

- How often do you attend the church services?
- The possible results as:

a) Regularly b) Occasionally c) Rarely D) Never

One long time church member, who attend service two or three each month, answered "Occasionally". Another person who attend service only at Christmas and Easter answered "Regularly". The responses did not provide the Council with useful information. This question suffer from measurement error.

How to avoid:-
Clear, Unambiguous questions would be asked so that the respondents are both capable of and motivated to answer correctly. Hence measurement Error is avoided.

3- Non-Response Error
Non-response error occurs when the survey fails to get a response to one, or possibly all, of the questions. A low response error, Under 50 to percent should be a response rate which will introduce a non-response error.

Sources:-
These errors occur when the survey fails to measure some of the units in the selected. Reason for this type of error may be that the respondent is unavailable or temporarily absent, the respondent is unable or refuses to participate in the survey. If a significance number of people do not

respond to a survey, the results may be biased. Another reason is that the characteristics of the non-respondent may differ from respondent.

Example:

The president of professional association decided to survey members about the Journals publication. She printed the questionnaire on the sheets of coloured paper to develop interest which then folded and mailed with bulk rate postage and does not follow-up contacts. Only 15% people who received the questionnaire responded. This survey suffered from potential non-response error.

How to avoid:-

→ This error can be avoided if everyone in the sample response to the survey or non-respondents are similar to respondents on characteristics of interest in a survey.

→ Design an interesting questionnaire

→ Design a questionnaire which is simple e.g. Complex questionnaire avoided.

4- Sampling Error:-

Sampling error occurs when researcher surveys only a subset of all people in the population instead of conducting a census. It can be measured by the difference of sample statistic and population parameter.

Example:-

A planner in a small community decided to interview a sample of residents about the attitudes they had towards the local school system. He developed a relatively complete list of population from utility companies and voter registration records. He then randomly chooses 50 residents for his survey. A sample size of 50 was not big enough to allow the planner to confidently generalize from his sample to the entire population.

How to avoid:-

→ Sampling error is a fact of life for those who conduct sample surveys. In other words it can never be completely avoided unless a census is done.

→ It can be minimize by calculating the better sample statistic.

→ Sampling error is something the researchers can control just by increasing sample size. Sampling error is inversly proportional to sample size.

Question:-

How to execute a successful survey or Discuss the main steps involved in a successful survey.

Ans:-

Following are the steps to execute a successful survey

1- Define your Purpose:-

Decide exactly what you want to learn from the survey. For instance, if you run a business, you may want to survey customers about their level of satisfaction with your service or product.

2- Keep it Short:-

There is a limit to how much time participants will devote your survey design, that people can complete in 5 minutes or less. If you have more questions than that save time for another survey.

3- Keep it Simple:-

Make sure that respondents will understand the questions. Dont use wrong and dont make the questionnaire too complex.

4- Put Demographic Question last:-

Keep information that is less crucial to your surveys towards the end. As people will lose interest in your survey.

5- Keep it Specific:-

Dont ask the open-ended question that will give you a wide range of answers.

It will make difficult to analyze the results
Questions should be either yes or no or
Multiple Choice.

6- Make it Consistent:-
If the first question ask your respondents to rate your customer survice on a scale of 1 to 5, with 5 being highly satisfactory, make sure that in subsequent questions, 5 always corresponds to being highly satisfactory.

7- Follow a logic:-
Make sure that one question leads naturally into another. usually the first question will be broader and the follow up ones will be more specific.

8- Do a test:-
Give the survey to a group of employes or customers. This will tell you how long it takes to complete, and also what if any question they find confusing or wrong.

9- Avoid weekends:-

10- Send reminders:-
If you email the survey, set a dead line to receive the results, few days before, send a reminder.

11- Entice:-
Give your customers a good reason to answer your survey, offer them a discount or give them a gift certificate.

12- Share:-
Last but not least, Share the results with Your customer and let them Know what action You will take.

Question:-
What is difference b/w Census and Sample Survey

Ans:-
- Census Survey:-
The census is the most complete source of information about the population that we have. Census survey is one in which we collect information from every person in the population.
Fedral, Provincial and local Government use Census survey.
Census results are goldmine of information

- Sample Survey:-
Survey. Sampling describes the process of selecting a Sample of element from a target population in order to Conduct a Survey. A Survey may refer to many different types of observation.
Sample Surveying is the art and Science of coming Close.

What is Survey?

A detailed study or inspection as by gathering information through observations, questionnaires etc. It is a general view; comprehensive study or examination. A study / survey may focus on opinions or factual information depending on its purpose but all survey involves administering questions to individual.

Survey assumes that answers to the questions lies in the present practice or opinion of the respondents. Statistical surveys are used to collect quantitative information in the fields of marketing, Political Polling, Social Science research etc.

Survey Characteristics:-

USAGE:- Surveys are widely used in Social Sciences.

Systematic: Follow specific procedures based on Survey science and scientific method.

Based on: Often presented as interviews or questionnaire.

Impartial Sampling: Units are selected from the population without Prejudice or preference and so as to be representative.

Data: Often quantitative, but can be qualitative.

Replicable: Other people using the same methods are likely to get essentially the same results.

Purposes of Survey :-

→ Information Gathering:
- i- Exploratory Study
- ii- Descriptive Study

i- Exploratory Study :-

The goal is to discover and explore phenomena e.g. would people be interested in our new product idea.

ii- Descriptive Study :-

The goal is to describe phenomena e.g. age, gender, Pay e.g. what features do buyers prefer in our product?

→ Theory Testing:
- i- Explanatory Study
- ii- Predictive Study

i- Explanatory Study:

The goal is to explain phenomena by looking at the relations b/w and patterns amongst variable e.g. cause and effect relationship i.e which of the two advertising campaign is more effective?

ii- Predictive Study:-

The goal is to be able to make accurate /useful predictions i.e What will happen?

- Advantages of Survey:

→ Surveys are flexible in a sense that a wide range of information can be collected. They can used to study attitudes, values, beliefs and Past behavior.

→ Access to wide range of Participants.

→ Potentially large amount of data

→ More ethical

→ They are relatively easy to administer

→ Because they are standardized they are relatively free from several types of errors.

- Disadvantages of Survey:

→ Lack of Control

→ Surveys are not appropriate for studying Complex social phenomena.

→ Data may be costly to obtain representative data.

→ Data may be Superficial

→ Self report data only.

Types of Survey :-

Following are the types of survey

i- Mail Surveys
ii- Teleplone Interviews
iii- Face to Face Interviews
iv- Drop - off Surveys.

i- Mail Surveys:-

Surveyors select their sample from a reasonably complete adress list of a population. Next, they may mail an advance notice letter, followed by a questionnaire (with a cover letter and stamped return envelope) to each member of the sample. Then, within about a week, they send a postcard reminder. Respondents complete the questionnaire and mail it back. People who donot return the questionnaire promptly can be contacted again, either by mail or telephone.

Advantages of Mail Surveys:-

1- The greatest strenght of mail surveys is that they require the least amount of resources.
2- Mail surveys are the easiest to do for people who have no experience and no professional help.
3- Another strenght of mail surveys is that they allow one to minimize sampling error at relatively low cost.
4- Mail surveys, especially when a survey is to be done locally, can provide a sense of privacy.
5- They are less sensitive to biases introduced by interviewers as well as to the tendency for respondents

to give answers they think the interviewer wants to hear.

Disadvantages of Mail Survey:-

1- The greatest weakness of mail surveys stems from their sensitivity to noncoverage error.

2- A second weakness of mail surveys is that some people are less likely to respond to the questionnaire than others.

3- Another weakness of mail surveys is that researchers have little control over what happen to the questionnaire after it is mailed.

4- Surveyors also cannot control whether mail questionnaires are filled out completely.

Mail Surveys are best Suited

→ Surveying people for whom a reliable address list is available and who are likely to respond accurately and completely in writing.

→ Surveys in which an immediate turnaround is not required.

→ Projects in which money, qualified staff, and Professional help. ~~are all~~

ii- Telephone Interviews:-

Surveyors select their sample from a telephone directory or other list. Alternatively, they can use one of several random-number techniques that have been developed to overcome list problems for general-public surveys. People in the sample are interviewed at the time of the first phone or at another, more convenient time. Interviewers record answers either on a survey form or, for some surveys, directly into a computer.

Advantages:-

1- The greatest strength of this method is its ability to produce results quickly.

2- In addition to quick turnaround, telephone surveys offer the advantage of greater interviewer control.

3- The cost of telephone surveys lies generally b/w that of face to face and mail surveys; its two main components are labour and long distance charges.

Disadvantages:-

1- Not all people have telephones. Hence a group of the population is automatically excluded from being surveyed.

2- Another reason some people are deterred from conducting telephone surveys is that telephone directories, the easiest lists from which to draw samples, are incomplete.

3- The first is that a knowledgeable supervisor is critical to success, especially when interviewers have not conducted surveys before.

4- Other problems with telephone surveys have to do with their sensitivity to measurement errors.

5- Many people give answers they think are socially acceptable, whether the question has to do with income, religious beliefs, etc.

Best Suited

- Time is less/limited
- members of the population are very likely to have telephones.
- Questions are relatively straight forward
- Experienced help is available
- Quick turnaround is important.

iii- Face-to-Face Interviews:-

If either a telephone or address list is available, Surveyors select their sample from the list and then contact each member to conduct the interview in person. If no suitable list is available, an area frame sampling technique is used. Interviewers conduct the survey by talking to respondents in person and recording the answer to each question on a survey form. If the respondent is not home when the interviewer arrives or if the interviewer is interrupted, he or she must make one or more additional visits to complete the Job.

Advantages:-

1- Surveying population for whom there is no list
2- Collecting information from people who are not likely to respond willingly or accurate by mail or telephone.
3- Complex Questionnaires
4- Well funded projects for which experienced interviewers and professional help are available.

Disadvantages:-

1- A good Supervisor is also a must for face to face surveys.
2- The temptation to cut costs in a face-to-face survey can be extremely high.
3- Unfortunately, cost-cutting carries a high price in terms of Error.
4- Face-to-face surveys are expensive but their strenghts should not be overlooked.

iv- Dropp-off Survey:-

One final method to consider is the drop-off survey, in which people deliver questionnaires by hand to households or businesses. Respondents complete the questionnaires on their own and either return them by mail or leave them out to be collected.

To make the (error) most of a drop-off survey. We recommend that surveyors leave questionnaires only with the intended respondents, rather then in mail boxes or with people who must convey the purpose of the survey to someone else. Personal contact enables to the surveyor to encourage respondents to complete the questionnaire. It also gives the survey a human face.

Drop of survey combine the low labor cost of mail surveys with the personal contact of face to face interviews.

Best Suited:-

→ Small community or neighborhood surveys in which respondents are not spread over a large area.

→ Relatively short and simple questionnaires

→ Projects with a small staff but relatively large sample size.

- **What is Sampling?**

The process of selecting a number of individuals for a study in such a way that the individuals represent the larger group from which they were selected.

- **Purpose of Sampling**

To gather data about the population in order to make an inference that can be generalized to the population.

- **Approaches to Sampling**

i- Probability Sampling:-

A probability sampling scheme is one in which every unit in the population has a chance of being selected in the sample, and this probability can be accurately determined.

ii- Non Probability Sampling:-

Is any sampling method where some elements of the population have no chance of selection, or where the probability of selection can't be accurately determined.